

## NAKUSP, BC, Canada

WMO: 712160

Lat: 50.2694N

Lon: 117.8172W

Elev: 1680

StdP: 13.83

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | 8.7        | 13.5 | 0.6                           | 6.0 | 10.9 | 5.5 | 7.8 | 15.2 | 12.1                  | 27.0 | 10.1 | 28.1 | 4.8                   | 100  | 0.507 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 25.5                   | 88.0            | 65.3 | 84.3 | 64.0 | 80.6 | 62.6 | 67.7                | 82.4 | 65.9 | 79.5 | 64.2 | 76.8 | 3.1                  | 260  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 63.0                            | 91.5 | 72.3 | 61.1 | 85.7 | 70.4 | 59.4 | 80.5 | 68.9 | 33.1          | 82.1 | 31.6 | 79.9 | 30.2 | 76.8 | 75.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 10.7          | 8.9  | 7.4 | 5.3                        | 94.8 | 5.7                | 4.1 | 1.3                                                | 97.7 | -2.1       | 100.0 | -5.3       | 102.3 | -9.4       | 105.3 |    |    |
| (5)               |      |     | 4.2                        | 70.8 | 5.6                | 2.8 | 0.2                                                | 72.8 | -3.1       | 74.5  | -6.3       | 76.0  | -10.4      | 78.1  |    |    |

## Monthly Climatic Design Conditions

|             |                                                                                  | Annual  | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|-------------|----------------------------------------------------------------------------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|             |                                                                                  | (d)     | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)         | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg   | 47.0  | 29.4  | 31.3  | 38.1  | 45.9  | 55.1  | 61.6  | 67.8  | 66.1  | 57.2  | 45.4  | 36.3  | 29.1 | (6)  |
| (7)         |                                                                                  | DBStd   | 14.91 | 7.39  | 6.64  | 5.37  | 4.66  | 5.76  | 5.48  | 5.60  | 5.34  | 5.52  | 5.13  | 5.80  | 6.94 | (7)  |
| (8)         |                                                                                  | HDD50   | 2909  | 638   | 523   | 368   | 138   | 18    | 0     | 0     | 0     | 6     | 159   | 412   | 647  | (8)  |
| (9)         |                                                                                  | HDD65   | 6805  | 1103  | 943   | 833   | 573   | 311   | 130   | 37    | 52    | 241   | 608   | 862   | 1112 | (9)  |
| (10)        |                                                                                  | CDD50   | 1829  | 0     | 0     | 0     | 15    | 174   | 349   | 552   | 500   | 223   | 16    | 0     | 0    | (10) |
| (11)        |                                                                                  | CDD65   | 249   | 0     | 0     | 0     | 0     | 3     | 29    | 123   | 87    | 7     | 0     | 0     | 0    | (11) |
| (12)        |                                                                                  | CDH74   | 2762  | 0     | 0     | 0     | 0     | 101   | 383   | 1307  | 880   | 91    | 0     | 0     | 0    | (12) |
| (13)        |                                                                                  | CDH80   | 897   | 0     | 0     | 0     | 0     | 14    | 102   | 489   | 274   | 18    | 0     | 0     | 0    | (13) |
| (14)        | Wind                                                                             | WSAvg   | 3.0   | 3.0   | 2.8   | 2.9   | 3.0   | 3.1   | 3.1   | 3.0   | 2.8   | 2.7   | 3.0   | 3.4   | 3.3  | (14) |
| (15)        | Precipitation                                                                    | PrecAvg | 32.8  | 3.9   | 2.6   | 2.2   | 2.0   | 2.5   | 3.2   | 2.6   | 2.3   | 2.5   | 2.5   | 3.6   | 3.9  | (15) |
| (16)        |                                                                                  | PrecMax | 43.5  | 6.6   | 6.3   | 4.4   | 4.3   | 4.1   | 6.0   | 5.8   | 5.3   | 5.0   | 6.4   | 9.2   | 7.4  | (16) |
| (17)        |                                                                                  | PrecMin | 23.8  | 0.7   | 0.6   | 0.1   | 0.6   | 0.4   | 1.0   | 0.3   | 0.2   | 0.3   | 0.4   | 0.9   | 1.9  | (17) |
| (18)        |                                                                                  | PrecStd | 4.9   | 1.7   | 1.3   | 0.9   | 0.9   | 1.0   | 1.2   | 1.5   | 1.4   | 1.3   | 1.5   | 1.8   | 1.5  | (18) |
| (19)        | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%    | DB    | 42.9  | 46.5  | 57.2  | 69.0  | 81.8  | 88.2  | 93.9  | 90.8  | 81.6  | 63.1  | 50.5  | 42.8 | (19) |
| MCWB        |                                                                                  |         | 40.0  | 41.3  | 45.6  | 53.2  | 59.4  | 65.1  | 67.5  | 66.8  | 61.7  | 54.5  | 47.2  | 39.3  | (20) |      |
| 2%          |                                                                                  | DB      | 39.8  | 42.6  | 52.2  | 63.3  | 76.6  | 82.6  | 89.4  | 86.3  | 76.1  | 58.9  | 47.2  | 40.0  | (21) |      |
|             |                                                                                  | MCWB    | 38.0  | 39.3  | 44.0  | 49.9  | 57.2  | 62.0  | 66.1  | 64.4  | 60.7  | 52.4  | 44.9  | 38.0  | (22) |      |
| 5%          |                                                                                  | DB      | 38.0  | 40.3  | 48.6  | 59.3  | 72.0  | 78.3  | 85.6  | 82.8  | 71.9  | 55.5  | 45.0  | 38.2  | (23) |      |
|             |                                                                                  | MCWB    | 36.8  | 38.0  | 42.0  | 47.9  | 55.8  | 60.7  | 65.1  | 63.7  | 59.3  | 50.9  | 43.3  | 36.8  | (24) |      |
| 10%         |                                                                                  | DB      | 36.7  | 38.4  | 45.7  | 55.8  | 67.6  | 74.0  | 81.8  | 79.0  | 67.7  | 52.8  | 43.2  | 36.6  | (25) |      |
|             |                                                                                  | MCWB    | 35.7  | 36.5  | 40.7  | 46.6  | 54.1  | 59.2  | 63.8  | 62.5  | 57.4  | 49.2  | 41.8  | 35.4  | (26) |      |
| (27)        | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%    | WB    | 40.9  | 42.6  | 47.6  | 54.6  | 62.0  | 67.2  | 70.9  | 69.5  | 64.5  | 56.7  | 48.6  | 41.0 | (27) |
| MCD B       |                                                                                  |         | 42.4  | 44.2  | 52.9  | 65.7  | 75.0  | 83.1  | 88.4  | 85.3  | 75.3  | 60.0  | 49.8  | 41.8  | (28) |      |
| 2%          |                                                                                  | WB      | 38.5  | 40.2  | 45.1  | 51.5  | 59.3  | 64.5  | 68.6  | 67.0  | 61.9  | 53.9  | 45.5  | 38.7  | (29) |      |
|             |                                                                                  | MCD B   | 39.3  | 41.8  | 49.5  | 60.3  | 71.8  | 77.7  | 84.1  | 80.7  | 72.4  | 57.0  | 46.4  | 39.4  | (30) |      |
| 5%          |                                                                                  | WB      | 37.2  | 38.5  | 43.3  | 49.4  | 57.3  | 62.4  | 66.7  | 65.3  | 60.2  | 51.7  | 43.7  | 37.1  | (31) |      |
|             |                                                                                  | MCD B   | 37.8  | 39.9  | 47.4  | 56.5  | 68.5  | 74.3  | 80.9  | 78.2  | 69.5  | 54.3  | 44.6  | 37.9  | (32) |      |
| 10%         |                                                                                  | WB      | 35.9  | 36.8  | 41.4  | 47.6  | 55.4  | 60.6  | 64.9  | 63.6  | 58.5  | 49.7  | 42.1  | 35.5  | (33) |      |
|             |                                                                                  | MCD B   | 36.5  | 37.9  | 44.9  | 54.0  | 65.5  | 71.0  | 78.1  | 75.6  | 66.1  | 52.2  | 43.1  | 36.3  | (34) |      |
| (35)        | Mean Daily<br>Temperature<br>Range                                               | 5% DB   | MDBR  | 6.8   | 10.5  | 14.5  | 19.4  | 22.7  | 22.0  | 25.5  | 24.6  | 19.8  | 12.8  | 7.6   | 6.1  | (35) |
| MCDBR       |                                                                                  |         | 7.5   | 11.7  | 18.7  | 26.1  | 31.4  | 30.9  | 32.6  | 30.9  | 26.2  | 16.9  | 10.0  | 7.4   | (36) |      |
| MCWBR       |                                                                                  |         | 6.0   | 8.4   | 11.4  | 13.9  | 14.8  | 13.8  | 13.9  | 13.2  | 12.8  | 10.5  | 7.5   | 6.0   | (37) |      |
| 5% WB       |                                                                                  | MCDBR   | 7.0   | 10.2  | 16.7  | 22.1  | 26.4  | 26.6  | 29.1  | 27.5  | 23.1  | 13.9  | 8.7   | 6.9   | (38) |      |
|             |                                                                                  | MCWBR   | 5.7   | 7.8   | 10.9  | 12.6  | 13.2  | 12.7  | 13.3  | 12.6  | 11.8  | 9.4   | 7.0   | 5.7   | (39) |      |
| (40)        | Clear-Sky<br>Solar<br>Irradiance                                                 | taub    | 0.286 | 0.288 | 0.309 | 0.358 | 0.385 | 0.404 | 0.375 | 0.389 | 0.353 | 0.334 | 0.299 | 0.274 | (40) |      |
| taud        |                                                                                  | 2.220   | 2.256 | 2.235 | 2.187 | 2.274 | 2.287 | 2.409 | 2.369 | 2.453 | 2.453 | 2.356 | 2.242 | (41)  |      |      |
| Ebn at Noon |                                                                                  | 234     | 269   | 283   | 279   | 276   | 271   | 278   | 268   | 267   | 251   | 230   | 220   | (42)  |      |      |
| Edh at Noon |                                                                                  | 23      | 29    | 35    | 42    | 40    | 40    | 35    | 35    | 29    | 24    | 20    | 19    | (43)  |      |      |
| (44)        | All-Sky Solar<br>Radiation                                                       | RadAvg  | 279   | 574   | 889   | 1386  | 1641  | 1695  | 1975  | 1691  | 1153  | 647   | 295   | 222   | (44) |      |
| (45)        |                                                                                  | RadStd  | 34    | 62    | 112   | 117   | 152   | 166   | 165   | 126   | 121   | 88    | 40    | 28    | (45) |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | -2.12 | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (15 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page